



Temperature dependence of the Brillouin linewidth in water

EDWARD FRY, JEFFREY KATZ, DAHE LIU & THOMAS WALTHER

To cite this article: EDWARD FRY, JEFFREY KATZ, DAHE LIU & THOMAS WALTHER (2002) Temperature dependence of the Brillouin linewidth in water, Journal of Modern Optics, 49:3-4, 411-418, DOI: [10.1080/09500340110088551](https://doi.org/10.1080/09500340110088551)

To link to this article: <https://doi.org/10.1080/09500340110088551>



Published online: 03 Dec 2010.



Submit your article to this journal [↗](#)



Article views: 369



View related articles [↗](#)



Citing articles: 8 View citing articles [↗](#)

Temperature dependence of the Brillouin linewidth in water

EDWARD S. FRY, JEFFREY KATZ, DAHE LIU[†] and
THOMAS WALTHER

Physics Department, Texas A&M University, College Station, TX
77843-4242, USA

(Received 15 April 2001; revision received 20 June 2001)

Abstract. In the frequency spectrum of light that is scattered in liquid water there is a central elastically scattered peak that is due mainly to scattering by suspended particles; and, there is a peak on each side of the central peak that is displaced by the Brillouin frequency shift. The Brillouin shift is a direct measure of sound speed. The linewidth of the Brillouin shifted lines is dependent on the bulk and shear viscosity of water as well as its density, thermal conductivity, and specific heat. The linewidth of the Brillouin peaks has been investigated in laboratory experiments over a range 1°C to 35°C. The frequency spectrum of back-scattered laser light was analysed using a scanning Fabry–Perot etalon. A strong dependence of the linewidth of the Brillouin shifted lines on temperature was found. In particular, for low temperatures in the range 10°C down to 1°C the linewidth shows an increase from 750 MHz to 1.4 GHz.

1. Background

Recently it has been shown that measurements of Brillouin shifts with pulsed lasers are feasible [1]. The prerequisite for such a task is a laser pulse with a Fourier transform limited linewidth. These measurements opened up the realm of remote sensing applications in the ocean utilizing a Brillouin LIDAR (light detection and ranging). Fry *et al.* performed detailed feasibility and accuracy studies for such a system [2].

For a fixed incident laser wavelength λ the Brillouin shift ν_B is proportional to the refractive index of water n and the sound speed v_s . The refractive index itself is dependent on the salinity S and temperature T . The dependence of the index of refraction on pressure will be neglected here, but could be easily included. The Brillouin shift is then expressed as:

$$\nu_B(S, T) = \frac{2n(S, T)}{\lambda} v_s(S, T) \sin(\theta/2), \quad (1)$$

where θ is the scattering angle ($\sin(\theta/2) = 1$ for 180° backscattering). Thus, there are five parameters: ν_B , v_s , n , S , and T . One of these is measured (ν_B) and there are three relations among them. These three relations consist of equation (1) and two empirical equations: one for the index of refraction [3] as a function of salinity S and temperature T , and the other for the speed of sound [4] as a function of T

[†] Also, Department of Physics, Beijing Normal University, Beijing 100875, PR China.

and S . To summarize, there are five parameters to be determined, whereas there is only one measurement and three relations. Thus, one more relation or parameter must be determined in order to arrive at a well-defined problem. Salinity was chosen as the additional parameter since data compiled between 1900 and 1990 by the National Oceanic and Atmospheric Administration [5] show that salinity values obtained from historical data are sufficiently constant to provide the required accuracy in T and S . Of course, salinity values differ from location to location and may vary seasonally, but for a fixed location and time of year, salinity data remain relatively constant. Fry and coworkers have found that if historical data are used for salinity, the accuracy limits on measurements of sound speed can be better than 0.3 ms^{-1} .

One purpose of this study is to investigate whether independent information on salinity and/or temperature could be measured so as to provide a fourth relation and thereby avoid reference to historical salinity data. To this end, the dependence of the Brillouin linewidth on temperature and salinity was studied for a temperature range of 1°C to 35°C and salinity values of 0‰ and 35‰.

2. Brillouin scattering

Brillouin scattering in a liquid has been studied by Mountain [6] and Fabelinskii [7] and was reviewed by Collins *et al.* [8]. The linewidth of the Brillouin peaks is given via the relation:

$$\delta = \frac{\Gamma}{2} \left(\frac{4\pi n}{\lambda} \sin(\theta/2) \right)^2, \quad (2)$$

where n is the index of refraction, λ is the wavelength of light, and the damping Γ is given by the relation

$$\Gamma = \frac{1}{\rho} \left(\frac{4}{3}\eta + \eta' + \frac{\kappa}{C_p}(\gamma - 1) \right)^2. \quad (3)$$

Here γ is the ratio of specific heats $\gamma = C_p/C_v$, ρ is the density of water, η is the shear viscosity, η' is the bulk viscosity and κ is the thermal conductivity of water. Data for thermal conductivity κ , density ρ and γ for water are available in the literature.

We are not aware of any reliable data giving the temperature dependence of the bulk viscosity of water in the frequency range of interest here. Specifically, the frequency of the relevant sound waves is identical to the Brillouin shift frequency ($\approx 7.5 \text{ GHz}$); and although the sound speed in water has been shown to be independent of frequency [8], this is not to be expected for the damping. Most measurements of bulk and shear viscosity in water have been obtained by absorption measurements of relatively low frequency acoustic waves and are not applicable here.

3. Experimental

The experimental setup for the Brillouin spectrum measurements has been described elsewhere [1] and is shown schematically in figure 1. Briefly, a frequency-doubled, injection-seeded, Q-switched, pulsed Nd:YAG laser provides

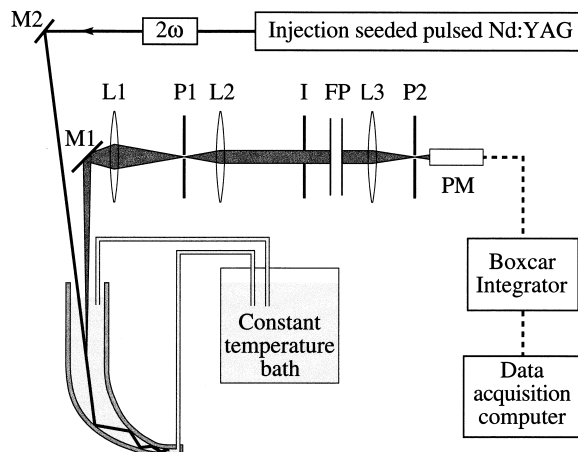


Figure 1. Schematic for the Brillouin spectrum measurement. L1, L2 and L3 are biconvex lenses with focal lengths 50 cm, 20 cm and 8 cm, respectively; P1 is a $10\ \mu\text{m}$ diameter pinhole; I is an iris; P2 is a 1 mm diameter pinhole; 2ω is a second-harmonic generating crystal; M1, M2 are mirrors; and PM is a photomultiplier tube. The horn-shaped water cell is wrapped with fibreglass insulation. The constant temperature bath maintains the water in the cell at a constant temperature that can be set from $\approx 0^\circ\text{C}$ to 35°C .

up to 200 mJ at 532 nm, with a repetition rate of 10 Hz. It operates at a single frequency and has stable Fourier transform limited pulses of 10 ns, i.e. a frequency bandwidth of about 50 MHz. A horn-shaped water cell is used to minimize back reflections. The water is circulated through a constant temperature bath that provides an adjustable temperature setting between 0°C and 50°C . The horn was wrapped with fibreglass insulation in order to cool and stabilize its temperature efficiently.

Back-scattered light is collected at an angle of 170° relative to the laser beam direction: several light baffles reduce parasitic back reflections. The acceptance angle of the optical system is $\approx 1^\circ$ and is determined by the focal length and the size of lens L1. The critical point to obtain good frequency discrimination is to provide a very well collimated beam to the input of the Fabry–Perot (FP) etalon. This is achieved via a spatial filter consisting of two lenses (L1 and L2), a $10\ \mu\text{m}$ diameter pinhole (P1), and an iris (I). To avoid background radiation, the output of the etalon is focused through pinhole P2 and onto the photomultiplier (PM); the overall receiver system is enclosed in a black box.

The transmission frequency of the Fabry–Perot was scanned at a rate of one FSR in two minutes via piezoelectric transducers that were driven by a Burleigh RC-42 ramp generator. The Fabry–Perot free spectral range (FSR) was calibrated by comparing the measurement of the Brillouin shift in units of scan voltage with calculated values of the Brillouin shift based on the known water temperature and salinity. The detector is a Hamamatsu R446 photomultiplier tube operating in the analogue mode. The signal is integrated and averaged over 10 laser shots by a gated integrator and boxcar averager. It is digitized and stored using a Labview program on a Mac II computer.

The water circulation was switched off during a measurement. The measurement intervals were typically on the order of a couple of minutes. No significant

temperature changes ($<1^\circ\text{C}$) were observed during this period. The measurements were performed by first adjusting the temperature settings for the heat exchanger. Then, after the temperature of the water in the cell had stabilized, a Brillouin spectrum was measured.

4. Results and discussion

Figures 2 and 3 show typical measured frequency spectra for water temperatures of 4°C (corresponding to a Brillouin shift of 7.14 GHz) and 19°C (corresponding to a Brillouin shift of 7.52 GHz), respectively. Each figure shows the data for a scan of the Fabry–Perot through three spectral ranges; they show the direct output of the boxcar averager with no additional processing. The Brillouin peaks are clearly visible. The central scattering peak in each FSR is due primarily to particles (hydrosols) in the water.

The figures also show the fit of a theoretical curve to the experimental data. The fitting function is based on a function given by the sum of a central Gaussian profile (laser line) with two Lorentzian profiles shifted by $\pm\nu_B$ with respect to it. The repetition of the signals due to the coverage of the scan over three free spectral ranges was taken into account in the fitting function by combining three of these sum functions that were shifted by successive free spectral ranges. The fit includes eight adjustable parameters: the Brillouin line shift ν_B , the free spectral range of the Fabry–Perot, the amplitude and linewidths of the Brillouin lines and of the central laser line, a nonlinearity factor to account for nonlinearities in the etalon

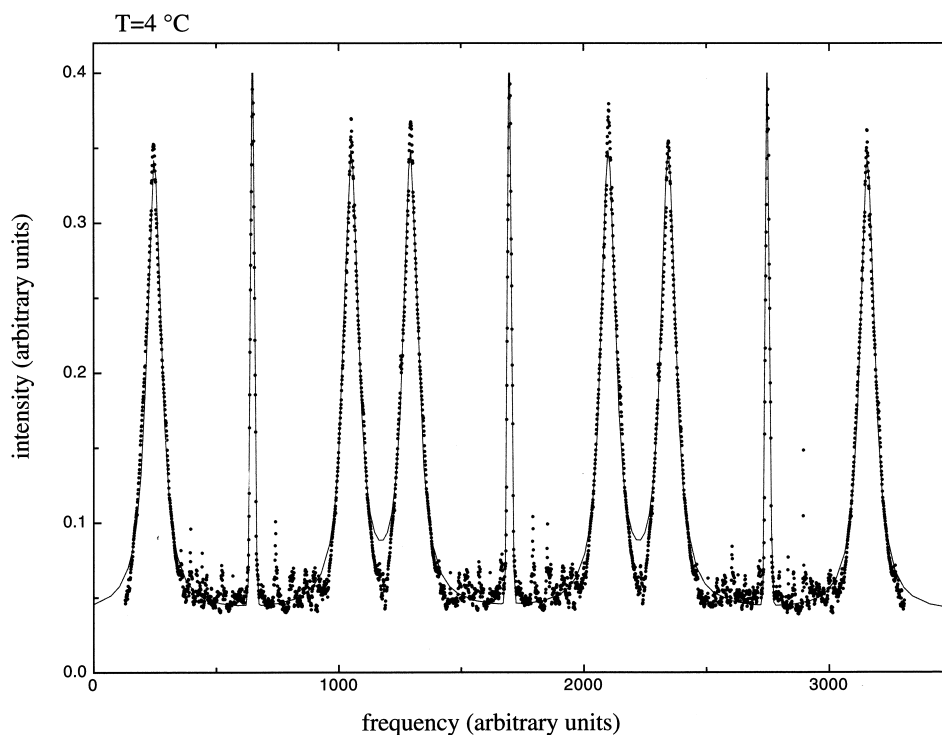


Figure 2. Experimental (·····) and best fit to the Brillouin spectrum (—) of water at 4°C .

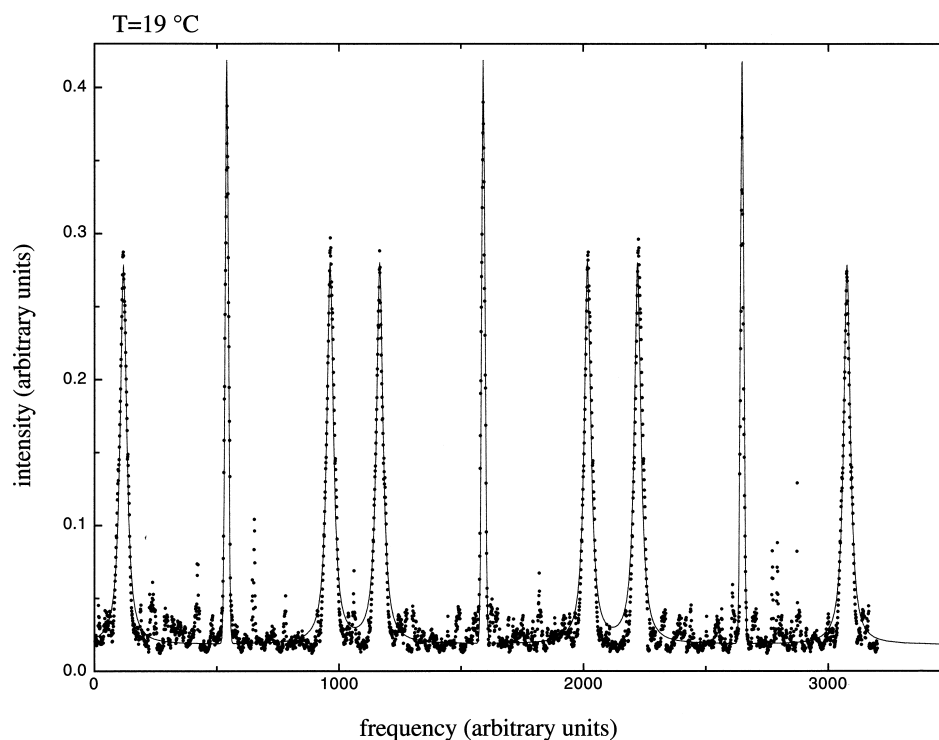


Figure 3. Experimental (·····) and best fit to the Brillouin spectrum (—) of water at 19°C. One clearly sees the difference in linewidth compared to the measurement of 4°C.

Table 1. Values of the Brillouin linewidth as a function of temperature and salinity measured at Texas A&M in 1998.

Temperature (°C)	Salinity (‰)	Brillouin linewidth (GHz)	Error (GHz)
2	0	1.45	0.09
3	0	1.26	0.04
4	0	1.25	0.04
5	0	1.25	0.04
8	0	0.99	0.05
10	0	0.79	0.07
15	0	0.59	0.16
19	0	0.54	0.08
35	0	0.54	0.08
4	35	1.1	0.1
10	35	0.78	0.09
30	35	0.41	0.13

scan, and an offset. The fits were performed on a PowerMac 8600 using the ProFit fitting program [9]. The fits were very good; only slight deviations at the foot of the peaks were observed. The absolute frequency calibration required for the linewidth determinations was obtained from the known Brillouin shifts at the respective temperatures; this shift can be calculated from equation (1). A clear dependence of Brillouin linewidth on temperature is found.

Measurements were made at several temperatures. Table 1 and figure 4 show all data collected at this time. There is a rapid increase in the linewidth of the

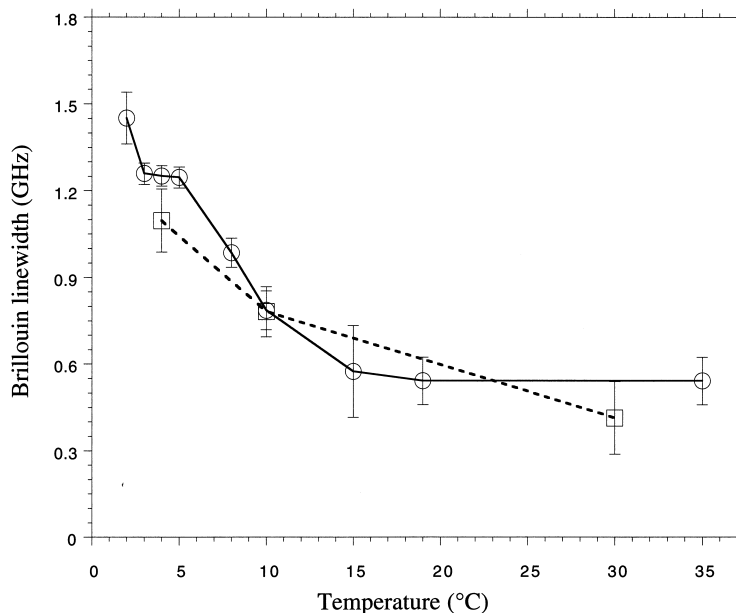


Figure 4. Data measured at Texas A&M in 1998 show the dependence of the Brillouin linewidth on temperature for 0‰ salinity ($\circ$ — $\circ$ — $\circ$) and 35‰ salinity ($\square$ — $\square$ — $\square$). The sharp rise of the linewidth towards lower temperatures is evident. Linewidth errors were determined from the fits to the data and its associated errors. In most cases the data for water with 35‰ salinity have larger errors, since the data have a larger noise component.

Brillouin peaks towards lower temperature. Most striking is the rapid increase for temperatures below 15°C.

One of the authors, Dahe Liu, repeated these measurements after returning to China. These more recent results are given in table 2 and shown in figure 5. For 0‰ salinity they are in excellent agreement with data given in table 1 and figure 4. For 35‰ salinity the observed linewidths tend to be generally larger; also, the studies were more extensive.

The measurements on the Brillouin shift in water with a salinity of 35‰ show much higher intensities in the central scattering peak due to higher hydrosol contamination. The spectra therefore have a larger noise component and the linewidth determinations have larger errors. Consequently, based on the present data, it is not possible to determine if salinity significantly influences Brillouin linewidth. However, the effect of salinity is clearly not as dramatic as that of temperature. More measurements to clarify this point are underway.

5. Prospects and conclusions

It has been shown that there is a strong dependence of Brillouin linewidth on temperature. The strong dependence at low temperatures indicates that temperature measurements might be feasible via linewidth measurements rather than from the Brillouin shift. Such data would provide a fifth parameter so that it would be possible to have independent determinations of all five parameters— S , T , ν_B , v_s and n . This independent determination of salinity is especially significant.

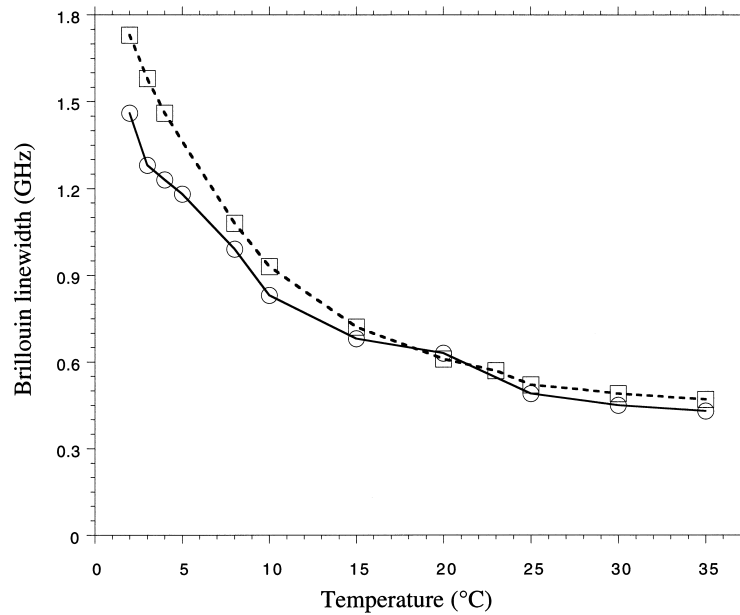


Figure 5. Data measured at Beijing Normal University in 2000 showing the dependence of the Brillouin linewidth on temperature for 0‰ salinity ($\circ$ — $\circ$ — $\circ$) and 35‰ salinity ($\square$ — $\square$ — $\square$).

Table 2. Values of the Brillouin linewidth as a function of temperature and salinity measured at Beijing Normal University in 2000.

Temperature (°C)	Salinity (‰)	Brillouin linewidth (GHz)
2	0	1.46
3	0	1.28
4	0	1.23
5	0	1.18
8	0	0.99
10	0	0.83
15	0	0.68
20	0	0.63
25	0	0.49
30	0	0.45
35	0	0.43
2	35	1.73
3	35	1.58
4	35	1.46
8	35	1.08
10	35	0.93
15	35	0.72
20	35	0.61
23	35	0.57
25	35	0.52
30	35	0.49
35	35	0.47

A measurement procedure for determination of the width of the Brillouin line can be analogous to that used for determination of lineshift. Specifically, the lineshift is accurately measured by employing an edge technique using absorption lines of $^{127}\text{I}_2$ and $^{129}\text{I}_2$ [1]. In this technique small frequency shifts are detected very sensitively by choosing the central laser frequency such that the Brillouin lines are located on the edges of absorption lines. Small variations in the shift translate into large intensity variations in the transmission of the absorption cell. The linewidth can be measured by employing two absorption lines that are located such that a Brillouin shifted frequency component is located on the red frequency side of one absorption line and on the blue edge of another absorption line. In this case the relative transmission signals can be used to find the width of the Brillouin shift. A Faraday Anomalous Dispersion Optical filter (FADOF) using potassium shows particular promise for such measurements [10].

Acknowledgments

The authors thank Professor George Kattawar for extensive, helpful discussions. We gratefully acknowledge support from the Robert A. Welch Foundation under Grant No. A-1218, from the Texas Advanced Research Program Grant No. 10366-0367-1999 and from the Office of Naval Research Grant No. N00014-96-1-0410.

References

- [1] EMERY, Y., and FRY, E. S., 1997, in *Ocean Optics XIII*, edited by S. G. Ackleson and R. Frouin, *Proc. SPIE*, **2963**, 210–215.
- [2] FRY, E. S., EMERY, Y., QUAN, X., and KATZ, J., 1997, *Appl. Opt.*, **36**, 6887–6894.
- [3] QUAN, X., and FRY, E. S., 1995, *Appl. Opt.*, **34**, 3477–3480.
- [4] DEL GROSSO, V. A., 1974, *J. Acoust. Soc. Am.*, **56**, 1084–1091.
- [5] 1993, Oceanographic station profile time series. CD-ROM, National Oceanographic Data Center (NODC), User Services Branch, NPAA/NESDIS E/OC21 (Washington, DC).
- [6] MOUNTAIN, R. D., 1966, *J. Res. National Bureau of Standards—Phys. Chem.*, **70A**, 207–220.
- [7] FABELINSKII, I. L., 1968, *Molecular Scattering of Light* (New York: Plenum Press), p. 622.
- [8] COLLINS, D. J., BELL, J. A., ZAONI, R., McDERMID, I. S., BRECKINRIDGE, J. B., and SEPULVEDA, C. A., 1984, in *Ocean Optics VII*, edited by M. A. Blizard, *Proc. SPIE*, **489**, 247–263.
- [9] 1996, Pro Fit 5.0 Quantum Soft (Zurich, Switzerland).
- [10] YANG, G., BILLMERS, R. I., HERCFELD, P. R., and CONTARINO, V. M., 1997, *Opt. Lett.*, **22**, 414–416.